



GREEN UNIVERSITY OF BANGLADESH (GUB)

Title

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A thesis submitted in the Department of X for the partial fulfillment of the degree of Y

Month, Year

Abstract

The abstract is typically a single paragraph. The abstract should be considered as an independent document, so that the abstract does not rely upon any material in the body of the report and, similarly, the body of the report does not rely upon any material in the abstract. The first sentence should clearly state the objective of the experiment. If the experiment is based upon a hypothesis, which is greatly preferred, the hypothesis should be stated and followed with statements describing its basis and evaluation. The subsequent sentences describe how the investigation was carried out. The following sentences describe, with as much precision as possible without being verbose, the results of the experiment. The final sentences describe the significance of the results and the impact of this work on the general field of study. Minimum 250 words

Acknowledgments

Provide a brief statement acknowledging the efforts of any participants or consultants who are not included as authors of the manuscript. State all of the funding sources for the work, ensuring that the statement adheres to the guidelines provided by the funding institution.

One can also mention and acknowledge supervisor, thesis or project mates, departmental lab facilities, scholarships or workshops, etc. However, acknowledgment must be in a single page.

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Chapter 1

Introduction

The introduction requires a short review of the literature pertaining to the research topic. The introduction is then best constructed as a descriptive funnel, starting with broad topics and slowly focusing on the work at hand [2]. Perhaps three to four paragraphs are needed. One approach may be to start with one or two paragraphs that introduce the reader to the general field of study. The subsequent paragraphs then describe how an aspect of this field could be improved [3]. The final paragraph is critical. It clearly states, most likely in the first sentence of the paragraph, what experimental question will be answered by the present study. The hypothesis is then stated [4, 5]. Next, briefly describe the approach that was taken to test the hypothesis. Finally, a summary sentence may be added stating how the answer of your question will contribute to the overall field of study.

1.1 Motivation

Start the section with your motivation on the topic. include reference here.

1.2 Aims and Objectives

Point out the aims of the research for example:

- To Identify the Challenges in Data Mining

- To establish a novel mathematical algorithm for X model
- To Develop a Novel System

1.3 Research Questions

2-3 solid research questions you would like to answer in your thesis

1.4 Research Contribution

What is new in this research and how people will get benefit from your work.

1.5 Thesis Outline

Give a brief outline of your thesis. For example:

- **Chapter 2: Literature Review** Tell us briefly what this chapter is about

Chapter 2

Literature Review

2.1 Introduction

In this chapter you need to study related works carried out previously. You must study at least 20-25 journal and conference articles and provide references. You need to provide critical review on the studied materials [6]. Just putting contents from other papers will not be sufficient. You must tell us the missing links in the studied papers and finally you need to provide a propose plan to address those missing links [7].

If you are providing any images or tables from a paper, ensure you provide accurate reference. see below.

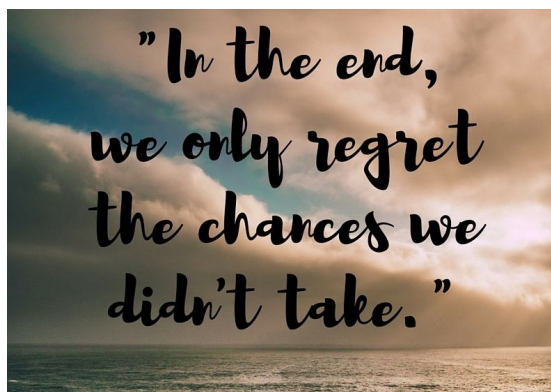


Figure 2.1: Figure name [1]

Background information is essential when it comes to writing thesis. It provides the

support that you need in order to make sure that your claim and or argument is backed up and that no one can say to you as the writer that you did not have enough evidence to support what you are claiming or which side you are for. When it comes to background information, become as knowledgeable about the subject as possible in order to help the researching process along and so you are not wasting valuable time with something that does not even matter.

2.2 Section

Add multiple sections for your research and complete the literature review section

2.3 Section

Add multiple sections for your research and complete the literature review section

2.3.1 Section

Add multiple subsections if required

2.3.2 Section

Add multiple subsections if required

2.4 Proposed Idea

Discuss your idea here in detail. You can present your idea with graphs, flowchart or any other means.

2.5 Conclusion

Here you provide your own critical review on studied paper.

Chapter 3

The Design Methods and Procedures

3.1 Introduction

This section should be a straightforward description of the methods used in your study. Each method should be described in a separate section. Begin, in a single section, with a statement of the materials used in the study, indicating the vendor and vendor contact information for each material. This information is critical so that readers have the capability to repeat the work in their own institutions. Next describe, in separate sections, each key procedure and technique used in the study. Keep explanations brief and concise. If a specific experimental design is utilized, describe this design in the second section of the Methods, after the materials section. Similarly, if a theoretical or modeling component is utilized, it should also be incorporated in the initial portion of the Methods. Finally, remember to describe the statistical analysis methods that were utilized to analyze the results, most likely in the final section of the Methods section. Although it is typically not recommended, the use of the passive voice is probably appropriate in the Methods section.

3.2 System Design

These are only examples of how one can add as many as sections and subsections needed according to the topic.



Figure 3.1: Figure (a)



Figure 3.2: Figure (b)

3.2.1 Subsection

You can use as many figures as required to explain your methodology.

- item 1
- item 2

3.2.2 Subsection

You may itemize your points using bullet marks.

- item 1
- item 2
- item 3
- item 4
- item 5

Subsubsection

You may use sub-subsections as required.

Subsubsection

You can fix height, width, position, etc., of the figure accordingly.



Figure 3.3: Figure name

Subsubsection

This is another sub-subsection.

3.2.3 Subsection

You can use tables to demonstrate your methodology and findings.

col 1	col 2	col 3	col 4	col 5

Table 3.1: Table name

3.2.4 Subsection

You can fix number of columns and rows of a table as required.

col 1	col 2	col 3

Table 3.2: Table

col 1	col 2	col 3	col 3	col 4	col 5

Table 3.3: Name of the table

3.3 Conclusion

Summarize the whole chapter.

Chapter 4

Evaluation of the Developed System

4.1 Introduction

4.2 Data Collection

Discuss experimental set up and environment installation needed for the simulation of your outcomes.

4.3 Data Analysis

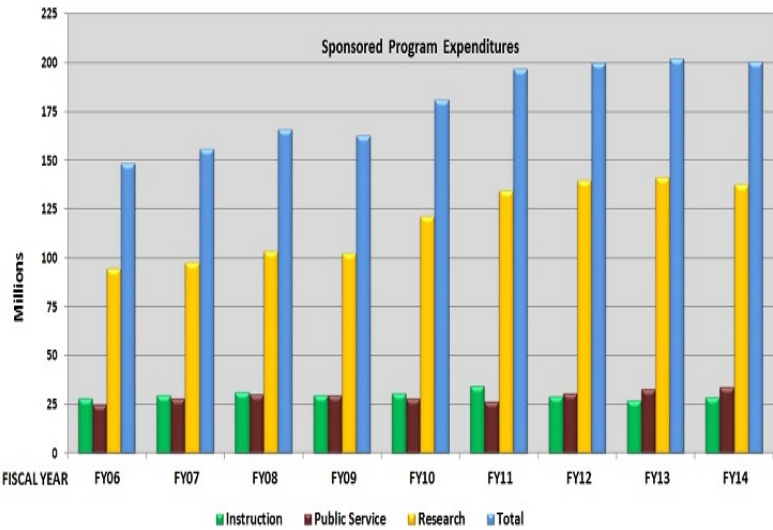


Figure 4.1: Name of figure

4.4 Optimization

Discuss the metrics or factors upon which you evaluated the performance of your work.



Figure 4.2: Figure name

4.4.1 Subsection

4.4.2 Subsection

4.4.3 Subsection

4.4.4 Subsection

The Results section presents the experimental data to the reader, and is not a place for discussion or interpretation of the data. The data itself should be presented in tables and figures. Introduce each group of tables and figures in a separate paragraph where the overall trends and data points of particular interest are noted. You may want to indicate the placement of a particular table or figure in the text. For experimental studies, key statistics such as the number of samples (n), the index of dispersion (SD, SEM), and the index of central tendency (mean, median or mode) must be stated. Include any statistical analysis that was performed, and make sure to indicate specific statistical data, such as p -values. Note that each table and figure in the paper must be referred to in the Results section. Be succinct.

4.5 Discussion

Discuss and summarize this chapter.

Chapter 5

Conclusion

Again, first introduce the work and then briefly state the major results. Then state the major points of the discussion. Finally, end with a statement of how this work contributes to the overall field of study

5.1 General Discussion

Summarize your complete research work here.

5.2 Limitation of the research

The discussion section, often the most difficult to write, should be relatively easy if the previous suggestions have been followed. In particular, look to the last paragraph of the introduction. If the work has characterized a phenomenon by studying specific effects, use the results to describe each effect in separate paragraphs. If the work has presented a hypothesis, use the results to construct a logical argument that supports or rejects your hypothesis. If the work has identified three main objectives for the work, use the results to address each of these objectives. A well-defined study that is described in the Introduction, along with supporting results that are presented in the Results section, should ease the construction of the Discussion section.

5.3 Practical Implications

Discuss the prospect and importance of your works practically.

5.4 Future Works

Propose the future research works related to your work so far. Also discuss the prospect of your work.

Bibliography

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- [4] K. Kambatla, G. Kollias, V. Kumar, A. Grama, Trends in big data analytics, *Journal of Parallel and Distributed Computing* 74 (7) (2014) 2561–2573.
- [5] W. Tan, M. B. Blake, I. Saleh, S. Dustdar, Social-network-sourced big data analytics, *IEEE Internet Computing* 17 (5) (2013) 62–69.
- [6] B. Hu, N. Carvalho, L. Laera, T. Matsutsuka, Towards big linked data: a large-scale, distributed semantic data storage, in: *Proceedings of the 14th International Conference on Information Integration and Web-based Applications & Services*, ACM, 2012, pp. 167–176.
- [7] D. Laney, 3d data management: controlling data volume, velocity and variety. gartner (2001).

List of Acronyms

WSN	Wireless Sensor Network
EH-WSN	Energy Harvesting Wireless Sensor Network
ORP	Optimal Relay Placement
SMRMC	Sustainable Minimum Relay Maximum Connectivity
MILP	Mixed Integer Linear Program
MATLAB	Matrix Laboratory

List of Symbols

β	Alpha
α	Beta
θ	Theta
γ	Gamma
ρ	Rho (Density)
π	Pi (3.1416)